

STAT 525

Chapter 11

Remedial Measures for Regression

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In this case the usual assumption $\text{Var}(\epsilon_i) = \sigma^2$ for every i is not met:

- Change distribution
- Change relationship between X and Y

If transformed, introduce non-linearity.

Problem 1: Unequal Var.

Unequal Error Variances

Error variance Depend on $-i^2$
 W : known matrix that weights each observation

- Consider $Y = X\beta + \varepsilon$ where $\sigma^2(\varepsilon) = W^{-1}$

$$\text{Var}(\varepsilon) = \sigma^2 W^{-1}$$

Case ①:
General Correlated Errors

- Potentially correlated errors and unequal variances

- Special case: $W = \text{diag}\{w_1, w_2, \dots, w_n\}$ ^{different values} ✓ If Diagonal of W has different values, then each obs. has different variance.

Case ②:
Errors Uncorrelated

- Heterogeneous variance or *heteroscedasticity* Problem!!

- Homogeneous variance or *homoscedasticity* if $w_1 = w_2 = \dots = w_n = 1/\sigma^2$

(OLS)

- Least square estimation still yields unbiased estimation, but is no longer optimal, and gives wrong uncertainty quantification ✗ You can still estimate β , but inference is unreliable.

• wrong SD
 • t-test, F-test invalid

$$E = (X^T X)^{-1} X^T Y = E(X^T X)^{-1} X^T X \beta = \beta$$

- Transformation of X or Y (e.g. Box-Cox) alone may unduly affect the relationship between X and Y
- Error variance is often a function of X or $E[Y]$

→ use WLS (weighted LS)

- Sometimes we try to stabilize the data by transforming Y or X but to avoid distorting the linear relationship, we can keep model linear and weight observations to their variance.

Transformation Approach

$$Y = XB + \epsilon, \text{ where } \text{Var}(\epsilon) = \sigma^2 W^{-1}$$

Step 1: By Pre-Multiplying by $W^{1/2}$ to get equal variance.

Step 2: Apply OLS

$\text{Var}(\epsilon_w) = W^{1/2} (\sigma^2 W^{-1}) W^{1/2} = \sigma^2 I$ • To make variance constant again, so we can apply LS.
Ex: GLS, WLS

- Consider a transformation based on a known W

$$W^{1/2} Y = W^{1/2} X \beta + W^{1/2} \epsilon$$

↓

$$Y_w = X_w \beta + \epsilon_w$$

- Can show $E(\epsilon_w) = 0$ and $\sigma^2(\epsilon_w) = I$
- **Generalized least squares:** apply the least squares method to

$$Y_w = X_w \beta + \epsilon_w$$

- It reduces to **weighted least squares** when W is a diagonal matrix

$w = \text{diag}(w_1, w_2, \dots, w_n)$ • Multiply each side by $\sqrt{w_i}$ gives WLS.

$w^{1/2} = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_n})$

- The transformation only requires that we know W up to some constant *only need to know relative weights*

Special Case: when W is Diagonal (no corr. between obs),
use WLS

Weighted Least Squares

Original OLS: $\sum_{i=1}^n (Y_i - X_i' \beta)^2$

- The least squares method minimizes

$$Q_w = (Y_w - X_w \beta)' (Y_w - X_w \beta) = (Y - X \beta)' W (Y - X \beta)$$

WLS - When $W = \text{diag}\{1/\sigma_1^2, 1/\sigma_2^2, \dots, 1/\sigma_n^2\}$,

$$Q_w = \sum_{i=1}^n \frac{1}{\sigma_i^2} (Y_i - X_i' \beta)^2$$

- (noisy)
• If obs. has $\uparrow \sigma^2$, we give $\downarrow$ weight ($\frac{1}{\sigma^2}$)
- More reliable obs. ($\downarrow \sigma^2$), we give $\uparrow$ influence ($\uparrow$ weight).
more precise

- By taking a derivative of Q_w , obtain normal equations:

$$(X_w' X_w) b = X_w' Y_w \quad \rightarrow \quad (X' W X) b = X' W Y$$

- Solution of the normal equations:

$$(X_w' X_w)^{-1} X_w' Y_w \quad \rightarrow \quad \underset{\text{WLS}}{b} = (X' W X)^{-1} X' W Y$$

– Can also be viewed as maximum likelihood estimator (MLE).

Feature	OLS	WLS
Unbiased estimates	Yes	Yes
Efficient (minimum variance)	No (if heteroscedastic)	Yes
Standard errors valid?	No	Yes
Hypothesis tests valid?	No	Yes

equal weights

If weights = 1
OLS

✓

different weights for each obs.

- In reality, we won't know the true σ^2 for each point. If we do, we could just use $w = 1/\sigma_i^2$ for each and be done.

Alternatives?

- Plot e_i v.s. X or $\hat{e}_i$ v.s. $\hat{Y}$
see if spread ↑↑↑ with X and Y .
- Regress $|e_i|$ on X to predict $|e_i|$ and to derive $w = 1/\sigma^2$
or e_i^2
- If in grouped data, compute e_i in each group, then assign $w = 1/\text{group var.}$

Weighted Least Squares (Continued)

- Easy to do in SAS using the **weight** option
- Must determine **optimal weights**
- Optimal weights $\propto 1/\text{variance}$ weight $(\frac{1}{\sigma^2})$
- Methods to determine weights, if no prior information of variance
 - Find relationship between the **absolute residual** and another variable and use this as a model for the standard deviation
 - Instead of the absolute residual, use the **squared residual** and find function for the variance
 - Use grouped data or approximately **grouped data** to estimate the variance

we don't need to know exact σ^2 , only how it changes with X .

Example Page 427

- Interested in the relationship between diastolic blood pressure and age
- Have measurements on 54 adult women
- Age range is 20 to 60 years old
- Issue:
 - Variability increases as the mean increases
 - Appears to be nice linear relationship
 - Don't want to transform X or Y and lose this
still linear, so we don't want to lose this. *✓ Use WLS*

ok
Age ↑, Blood Pressure ↑, but
Variability also ↑↑
not ok.

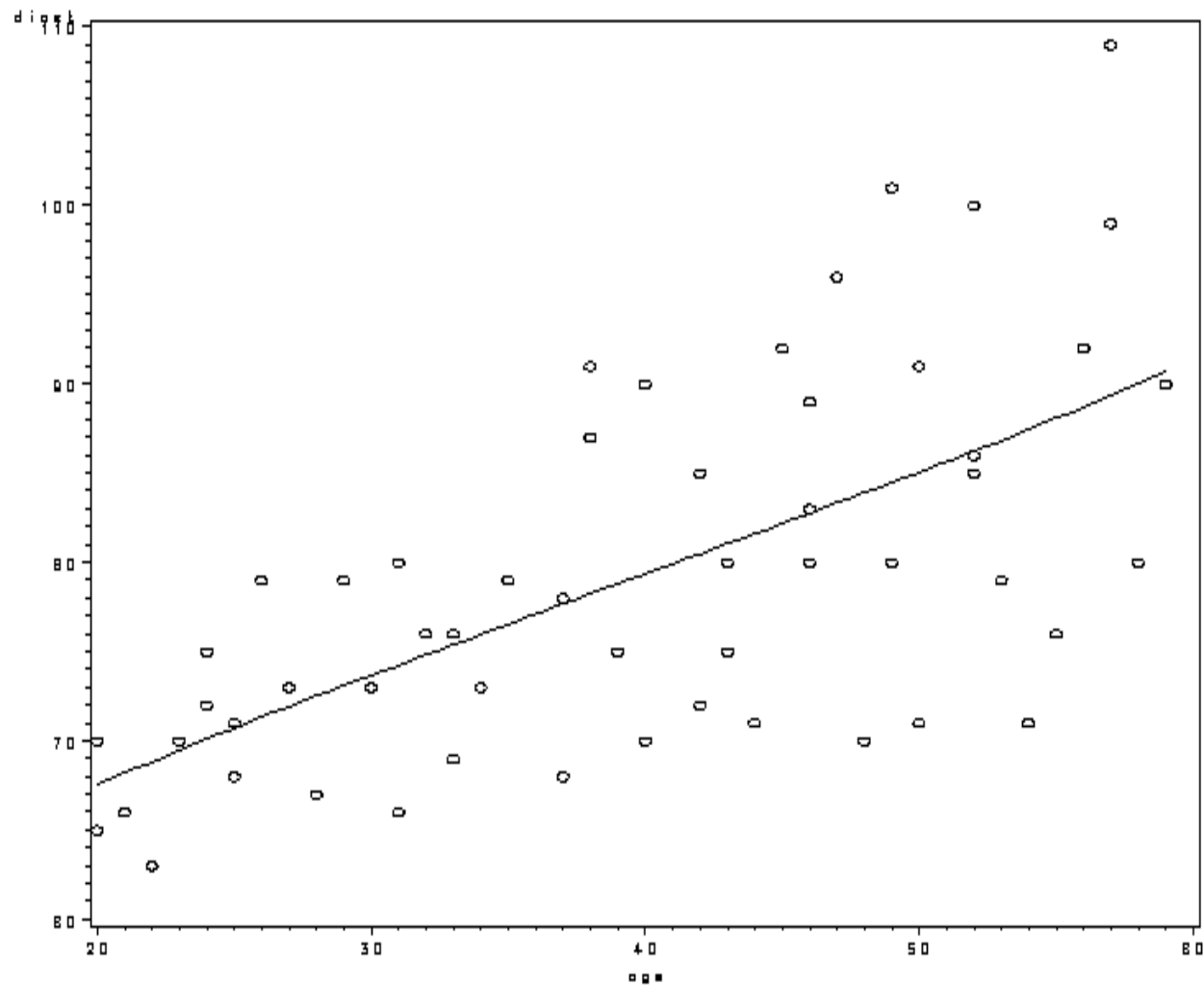
```
data a1;  
  infile 'U:\.www\datasets525\ch11ta01.txt';  
  input age diast;  
run; quit;
```

```

/* Scatter Plot */
proc sort data=a1; by age;
symbol v=circle i=sm70;
proc gplot data=a1;
    plot diast*age/frame;
run;

```

Heteroscedasticity:
 ↑ spread at ↑ Ages



- Image above motivates the use of WLS, but we still need to fit a regular OLS first to diagnose.
 ↓
 (to get fitted line and e_i)

```
/* Fit a Regular Regression */
proc reg data=a1;
    model diast=age;
    output out=a2 r=resid;
run;
```

OLS

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	2374.96833	2374.96833	35.79	<.0001
Error	52	3450.36501	66.35317		
Corrected Total	53	5825.33333			

Root MSE	8.14575	R-Square	0.4077	Moderate
Dependent Mean	79.11111	Adj R-Sq	0.3963	
Coeff Var	10.29659			

Parameter Estimates

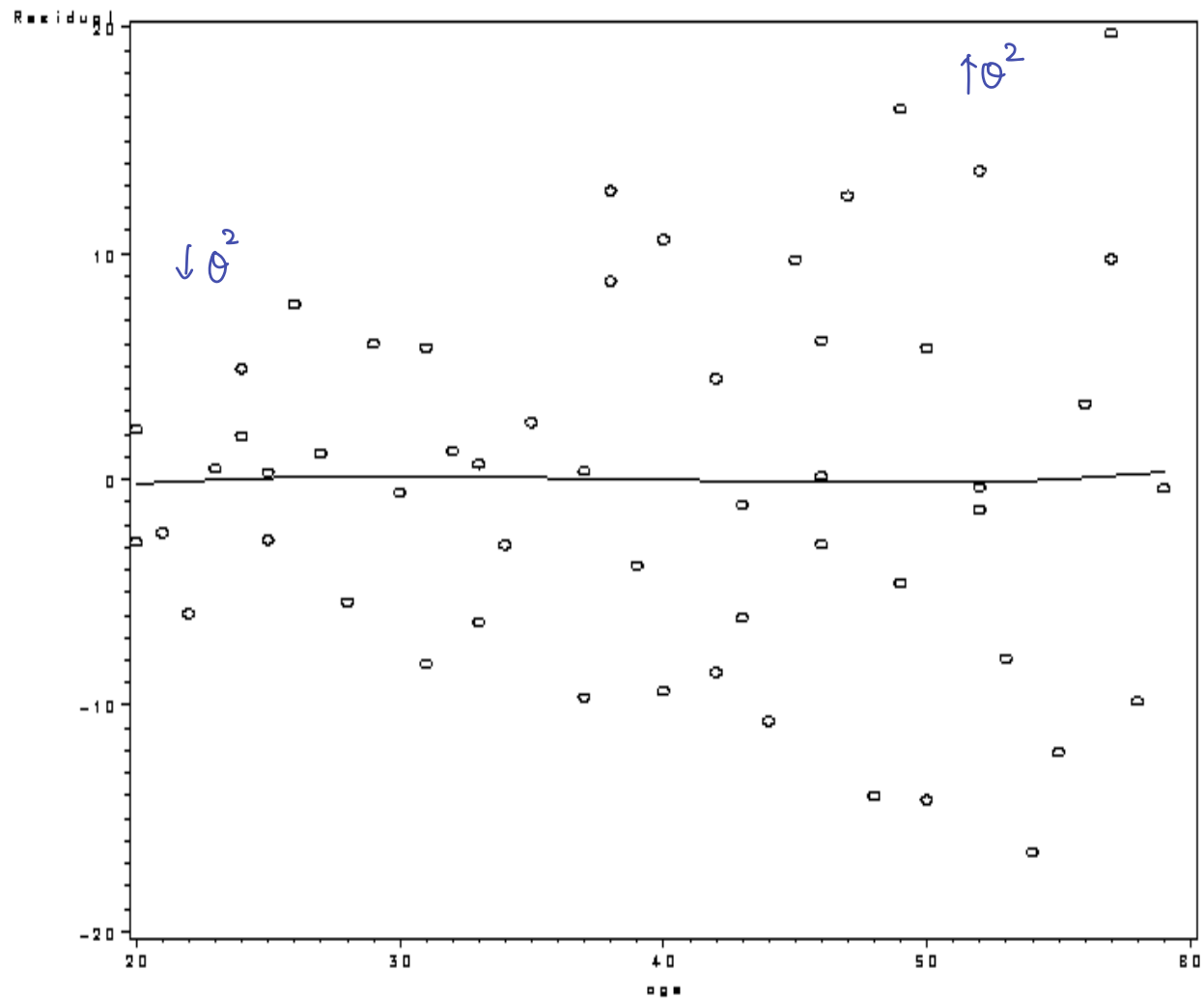
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	56.15693	3.99367	14.06	<.0001
age	1	0.58003	0.09695	5.98	<.0001

✓, but doesn't guarantee homoscedasticity, so we check e_i next: ↓


```

/* Residual Plot */
proc gplot data=a2;
  plot resid*age;
run; quit;

```



$\sigma^2 \uparrow$ with Age.

- The error variance increases as age increases

```
/* Find Pattern of Residuals vs Age */
```

```
data a2;
```

```
set a2;
```

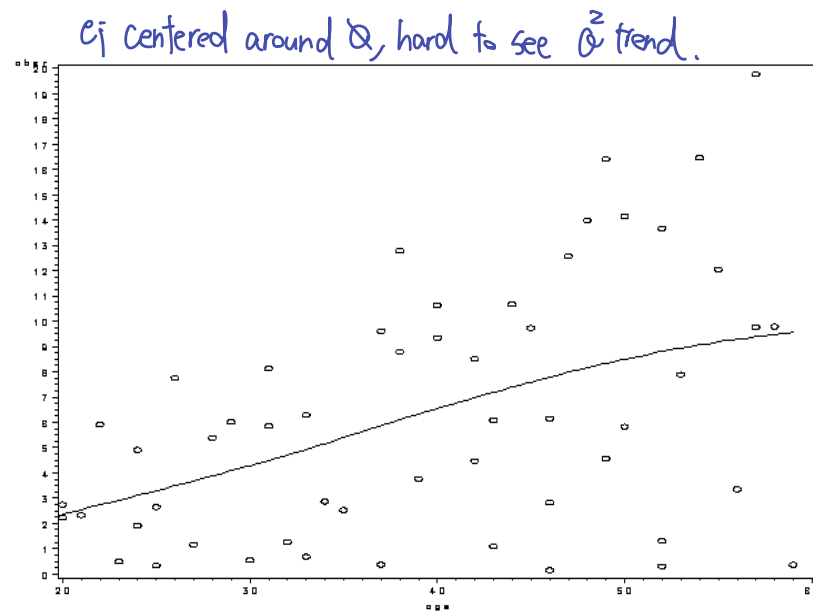
```
absr=abs(resid);
```

```
sqrr=resid*resid;
```

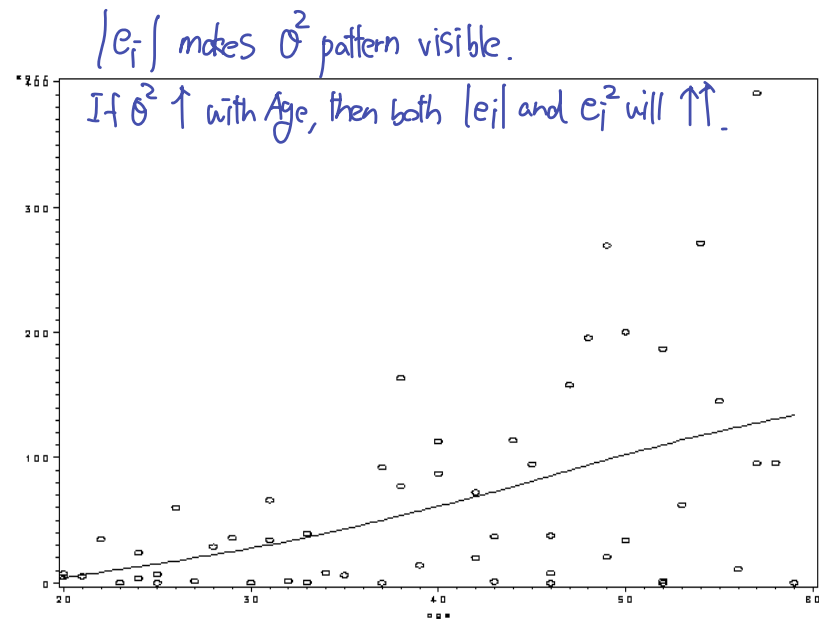
```
proc gplot data=a2;
```

```
plot (resid absr sqrr)*age;
```

```
run;
```



abs(Residual) vs. Age



(Absolute)

Residual² vs. Age

Construction of Weights

- Assume $\text{abs}(\text{res})$ is linearly related to age
- Fit least squares model and estimate σ_i

```
proc reg data=a2;  
  model absr=age;  
  output out=a3 p=shat;  
run;
```

- Take Weight as $w_i = 1/\hat{\sigma}_i^2$

```
data a3; set a3;  
  wt=1/(shat*shat);  
  
proc reg data=a3;  
  model diast=age / clb;  
  weight wt;  
run; quit;
```

⊗ not e_i^2 because squared
can be too noisy + unclear.

↖ $|e_j|$

$w = 1/\hat{\sigma}^2$, estimate $\hat{\sigma}^2$ using absolute e_j
from OLS model.

✓ Compare OLS vs WLS

WLS

Statistic	OLS (Slide 11-7)	WLS (Slide 11-11)	Change
β_1 (Age slope)	0.5800	0.5963	Slight increase
Std Error of β_1	0.09695	0.07924	✓ Smaller SE (good)
t-statistic for β_1	5.98	7.53	✓ More significant now
Root MSE	8.15	1.21	✓ Much improved
R^2	0.41	0.52	Not meaningful in WLS

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	83.34082	83.34082	56.64	<.0001
Error	52	76.51351	1.47141		
Corrected Total	53	159.85432			

Root MSE	1.21302	R-Square	0.5214
Dependent Mean	73.55134	Adj R-Sq	0.5122
Coeff Var	1.64921		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	55.56577	2.52092	22.04	<.0001	50.50718 60.62436
age	1	0.59634	<u>0.07924</u>	7.53	<.0001	0.43734 0.75534

Lower SE

- Not much difference in the estimates but a slight reduction in the standard deviations. Should not interpret R^2 in this situation.

Modifies OLS to stabilize model when predictors are correlated.

Ridge Regression as Multicollinearity Remedy

- Modification of least squares that overcomes multicollinearity problem
- Recall least squares suffers because $(X'X)$ is almost singular thereby resulting in highly unstable parameter estimates
- Ridge regression results in biased but more stable estimates
- After standardizing data, we consider the correlation transformation so the normal equations are given by $r_{XX}b = r_{YX}$. Original
Since r_{XX} difficult to invert, we add a bias constant, c .

$$b^R = (r_{XX} + cI)^{-1}r_{YX}$$

Adds a positive constant (c) to diagonal to make matrix invertible.

We then transform it back to coefficient estimators for the original data.

$$\begin{aligned} (X'X)\hat{\beta} &= X'Y \\ \downarrow \\ (X'X + cI)\hat{\beta}_R &= X'Y \end{aligned}$$

✓ Key trade-off

OLS	Ridge Regression
Unbiased	✗ Biased
High variance when multicollinearity exists	✓ Lower variance, more stable
Might not work well	✓ Better predictive accuracy

Ridge accepts some bias in exchange for stability.

Choice of c

- Key to approach is choice of c
- Common to use the *ridge trace* and VIF's

Method ① – Plot showing how each $\hat{\beta}$ change as we $\uparrow c$
 Ridge trace: simultaneous plot of $p - 1$ parameter estimates for different values of $c \geq 0$. Curves may fluctuate widely when c close to zero but eventually stabilize and slowly converge to 0. β_0

Method ② – VIF's tend to fall quickly as c moves away from zero and then change only moderately after that

X-axis: c (from 0 upward)
 Y-axis: value for each $\hat{\beta}$
 (one per predictor)

- Choose c where things tend to "stabilize"
 Choose where ridge trace levels off

- MODEL statement of PROC REG has option **ridge=c**

- $c = 0 \rightarrow$ OLS; unstable est./ $\uparrow$ SE
- $c \downarrow \downarrow \downarrow \rightarrow$ unstable
- c moderate $\rightarrow$ Best; stable
- $c \uparrow \uparrow \uparrow \rightarrow \hat{\beta}$'s biased toward 0

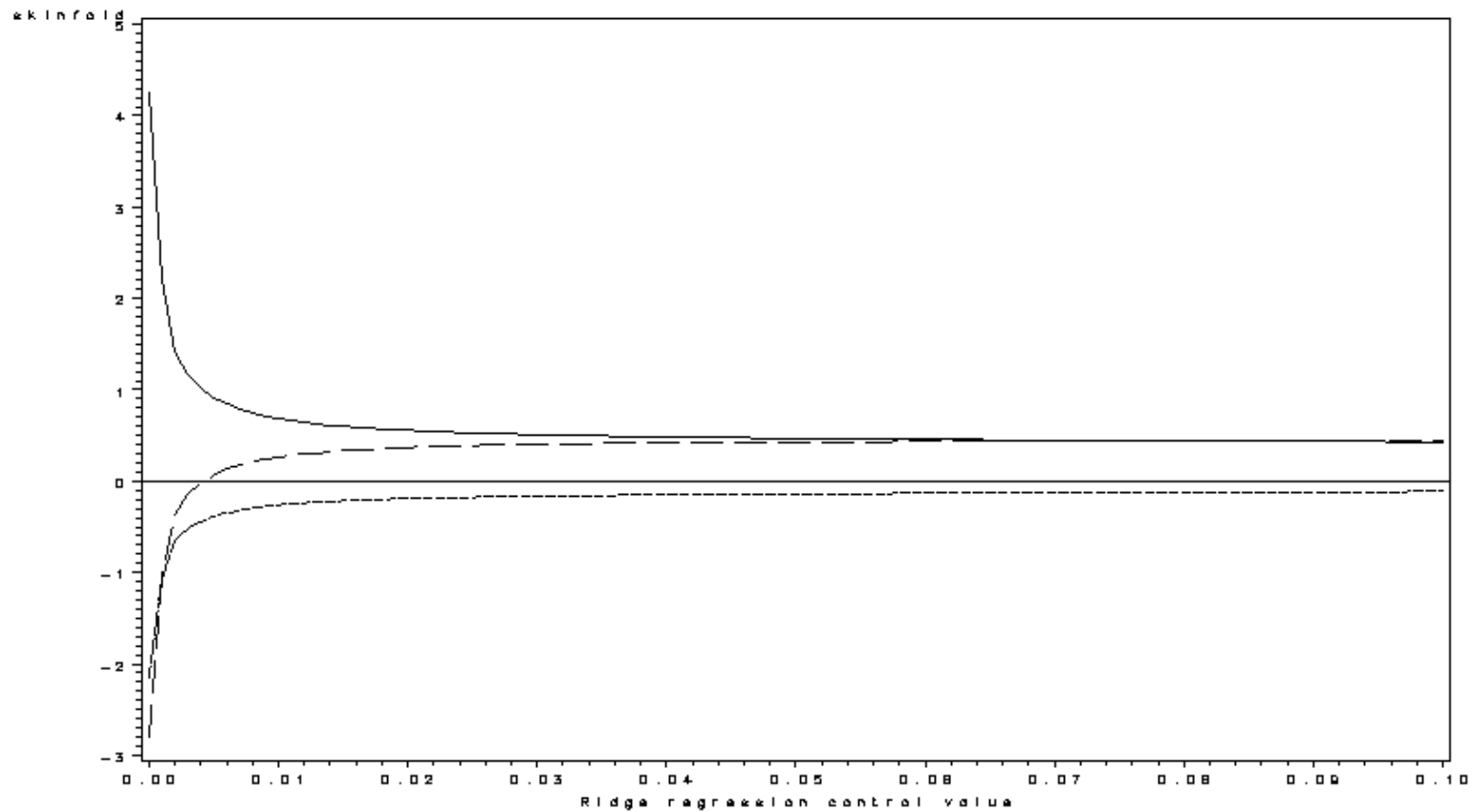
SAS Commands

```
data a1;
  infile 'U:\.www\datasets525\ch07ta01.txt';
  input skinfold thigh midarm fat;

/* ridge estimation are stored in the dataset designated by outest option */
proc reg data=a1 outest=b;
  model fat=skinfold thigh midarm /ridge=0 to .1 by .001;
run;

symbol1 v='' i=sm5 l=1;
symbol2 v='' i=sm5 l=2;
symbol3 v='' i=sm5 l=3;
proc gplot;
  plot (skinfold thigh midarm)*_ridge_ / overlay vref=0;
run; quit;
```

Ridge Trace



```
/* Another Way to get the Ridge Trace Plot */  
proc reg data=a1 outest=b;  
  model fat=skinfold thigh midarm /ridge=0 to .1 by .001;  
  plot / ridgeplot vref=0;  
run;
```


Robust Regression with Influential Cases

- Want procedure that is not sensitive to outliers
- Focus on parameters which minimizes
 - sum of absolute values of residuals (LAR: Least Absolute Residuals)
 - median of the squares of residuals (LMS: Least Median of Squares)
- Could also consider iterating through weighted LS where the residual value is used to determine the weight (IRLS)
Iteratively Reweighted Least Squares
- See pages 439-449 for more details
- Both robust and ridge regression are limited by more difficult assessments of precision (i.e., standard errors). Bootstrapping is often used.

(IRLS)

Instead of OLS/WLS, we fit model that
changes weights during fitting.

Iteratively Reweighted Least Squares Using PROC NLIN

- PROC NLIN allows to define weights as a function

```
/* NOHALV: removes the restriction that the objective value must  
decrease at every iteration */
```

```
PROC NLIN DATA=a1 NOHALV;
```

```
PARMS b0=0 b1=0;
```

```
MODEL diast = b0+b1*age;
```

```
resid = diast-MODEL.diast;
```

```
_WEIGHT_ = 1/(resid**2);
```

```
RUN; QUIT;
```

re-weight obs. based on their e_i

The NLIN Procedure

Source	DF	Sum of Squares	Mean Square	F Value	Approx Pr > F
Model	1	1.877E23	1.877E23	1.81E23	<.0001
Error	52	54.0000	1.0385		
Corrected Total	53	1.877E23			

Parameter	Estimate	Approx Std Error	Approximate 95% Confidence Limits		
b0	56.8462	3.8E-11	56.8462	56.8462	
b1	0.5385	1.27E-12	0.5385	0.5385	

Approximate Correlation Matrix

	b0	b1
b0	1.0000000	-0.9999964
b1	-0.9999964	1.0000000

When to use IRLS:

- few extreme data points you don't want to delete, but also don't want them dominating the model.
- You still believe general relationship is valid.

Nonparametric Regression

- Helpful in exploring the nature of the response function
- $i=sm\#\#$ is one such approach (Spline)
- All version have some sort of smoothing, via local averaging or global smooth basis functions
- See pages 449-453 for more details
- Interesting theory but confidence intervals and significant tests not fully developed

Machine Learning algorithms

Flexible modeling with few inference tools. Usually requires iterative optimization algorithms.

Regression Trees

- Piecewise constant regression function
- Basically partition the X space into rectangles
- Predicted value is mean of responses in rectangle
- Minimize SSE via greedy search (sequentially partitioning)
- Trade off between minimizing SSE and complexity

Instead of drawing a smooth curve, they split data into groups.



Neural Networks

- $y = B_1 \circ \sigma \circ B_2 \circ \sigma \dots B_m \circ x$
- Combination of nonlinear activation function σ and linear mapping B_i 's.
- Estimate B_i 's via minimizing squared loss
- Highly non-convex optimization task
- Foundation of deep learning

great for predictions but not
for traditional inference.

Evaluating Precision in Nonstandard Situations

- Standard methods for evaluating the precision of sample estimates may not be available or may only be approximately applicable when the sample size is large
 - Ridge regression (cons: biased coefficients, complicated var.)
 - Robust regression (cons: weights depend on e_i)
- **Bootstrapping** provides estimates of the precision of sample estimates
 - **Very** important theoretical development that has had a major impact on applied statistics
 - Resampling idea: use the sample to generate a “population” and generate new “samples” from such “population”
 - Use the pool of estimates from new “samples” to profile sample estimates (i.e., parameters of “population”)

Resampling Residuals (Fixed X sampling)

- Take the residuals $\{e_1, e_2, \dots, e_n\}$ as the “population” of error term ϵ
 - Sample ϵ_i^* from $\{e_1, e_2, \dots, e_n\}$
 - Let $Y_i^* = b_0 + b_1 X_i + \epsilon_i^*$
 - In the new “sample”, the i -th observation is (X_i, Y_i^*)
 - Assume constant error variances
- Useful when
 - errors have unknown distribution (but constant variance), and/or
 - want to preserve predictors
- Examples of use:
 - Ridge regression
- May sample ϵ^* from a “parametric population” of residuals

Resampling Pairs (Random X Sampling)

- Useful when
 - Doubt about the adequacy of the regression function being fitted
 - Unequal error variances
 - Predictor variables cannot be regarded as fixed
- Take $\{(X_i, Y_i) : i = 1, 2, \dots, n\}$ as the “population” of (X, Y)
 - For the new “sample”, the i -th observation (X_i^*, Y_i^*) is sampled from the “population” $\{(X_i, Y_i) : i = 1, 2, \dots, n\}$
- Examples of use:
 - Weighted regression

 Quick comparison table

Feature	Resampling Residuals	Resampling Pairs 
Keep X fixed?	✓ Yes	✗ No (resamples X and Y together)
Assumes model form is correct?	✓ Yes	✗ No
Handles heteroscedasticity well?	✗ Not great	✓ Yes
Use for experiments?	✓ Yes	✗ No
Use for observational data?	⚠ Maybe	✓ Best choice
What's resampled?	Errors (residuals)	Whole data points (rows)

Bootstrap Inference

- A total of B new “samples” can be generated, with each new “sample” providing an estimate of the parameter, say $b_1^{(k)}$ for β_1 from k -th new “sample”
 - Use $\{b_1^{(k)} : k = 1, 2, \dots, B\}$ to understand the population property of b_1

- Bias

$$Bias = E\{b_1\} - \beta_1 \implies \widehat{Bias}_{boot} = \bar{b}_1^* - b_1$$

$$\text{where } \bar{b}_1^* = \sum_{k=1}^B b_1^{(k)} / B$$

- Variance

$$Var = E\{(b_1 - E\{b_1\})^2\} \implies \widehat{Var}_{boot} = \frac{1}{B} \sum_{k=1}^B (b_1^{(k)} - \bar{b}_1^*)^2$$

Bootstrap Confidence intervals

- CI for β_1 with unbiased estimator b_1 *Percentile Method*

$$(b_1^*(\alpha/2), b_1^*(1 - \alpha/2))$$

- $b_1^*(\alpha/2)$ is the $(\alpha/2) \times 100$ percentile of $\{b_1^{(k)} : k = 1, 2, \dots, B\}$
- $b_1^*(1 - \alpha/2)$ is the $(1 - \alpha/2) \times 100$ percentile of $\{b_1^{(k)} : k = 1, 2, \dots, B\}$

- *Reflection Method:* CI for β_1 with biased estimator b_1

$$(b_1 - d_2, b_1 + d_1)$$

- $d_1 = b_1 - b_1^*(\alpha/2)$
- $d_2 = b_1^*(1 - \alpha/2) - b_1$

Example: Typographical Errors (4.12 on Page 173)

```
options nocenter; goptions colors=(none);

/* ----Read in initial data set and fit the model----*/
data a1;
    infile 'U:\.www\datasets525\CH04PR12.txt';
    input y x;

proc reg;
    model y=x / noint clb;
    output out=a2 p=pred r=res;
run;
```

Output from Proc Reg

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
x	1	18.02830	0.07948	226.82	<.0001

Variable	DF	95% Confidence Limits
x	1	17.85336 18.20325

```

/* Resample Residuals */

/* Create a data set that contains 1000 copies of the predictor
   variable and associated fitted value from the regression */
data pred; set a2;
  do sample=1 to 1000;
    output;
    keep sample x pred;
  end;
proc sort; by sample;
run;

/* Randomly sample (with replacement) the residuals => 1000 copies */
/* PROC SURVEYSELECT: Selecting random samples */
/* METHOD=URS: Select with equal probability & with replacement */
/* SAMPSIZE: Specifies the sample size */
/* REP: Number of samples (i.e., datasets) */
/* OUTHITS: Includes a separate observation in the output dataset for each
           selection when the same unit is selected more than once */
/* ID: variables to be included in the output dataset, all by default */
proc surveyselect data=a2 method=urs sampsize=12 rep=1000 outhits out=res;
  id res;
run;

```

```

/* Merge the fitted values and the residuals, and generate new y */
data new;
    merge pred res;
    ynew = pred + res;
run;

/* Perform regression on each new sample and store parameter estimate
   results in a dataset called parm */
/* The ods listing turns off the output going into the output window */
ods listing close;
proc reg data=new;
    model ynew=x / noint;
    by sample;
ods output ParameterEstimates=parm;
ods listing;

/* Generate histogram and approximate the density */
/* PCTLPRE: Specifies prefixes to create variables names for PCTLPTS */
/* PCTLPTS: Specifies percentiles to compute */
proc univariate noprint data=parm;
    var Estimate;
    histogram Estimate / kernel ;
    output out=a4 mean=bmean std=bsterr pctlpre=perc_ pctlpts=2.5,5,95,97.5;

proc print data=a4; run; quit;

```

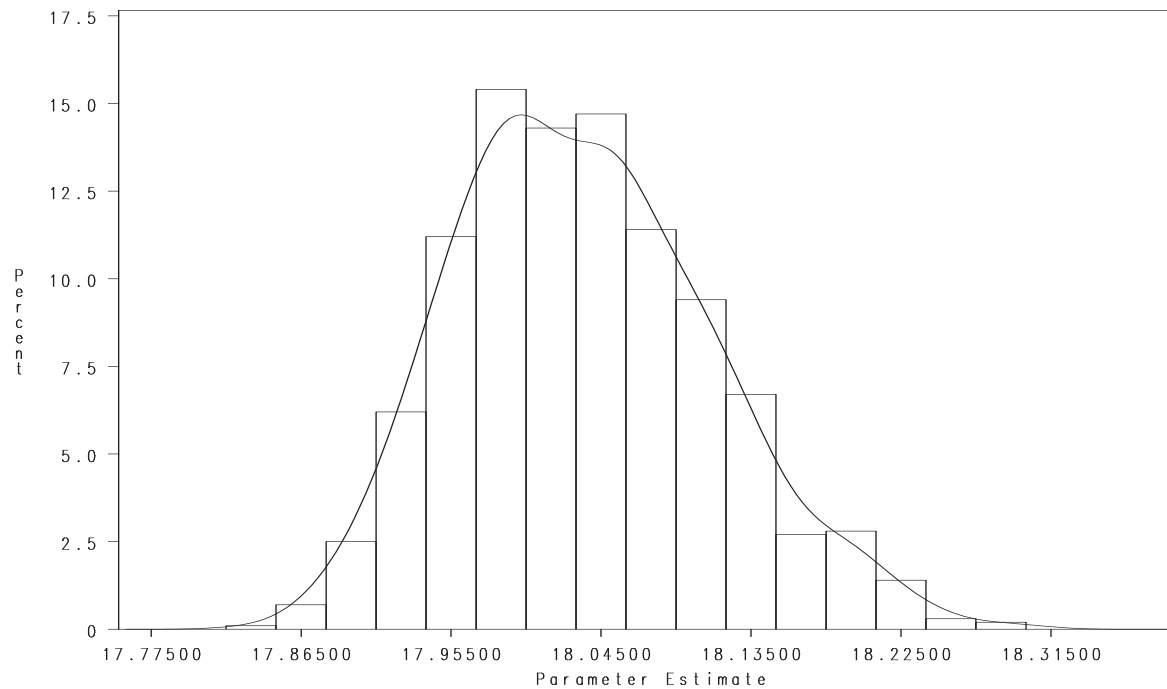
Results from Bootstrapping

Obs	bmean	bsterr	perc_2_5	perc_5	perc_95	perc_97_5
1	18.0348	0.076574	17.9038	17.9200	18.1747	18.1986

Bias = $18.0348 - 18.0283 = 0.0065$ (quite small)

Percentile : (17.9038, 18.1986)

Reflection : (17.8580, 18.1475)



Histogram of $\{b_1^{(k)} : k = 1, 2, \dots, 1000\}$

- The way to resample in the example is the easiest to implement
- But it is not a computationally efficient way to do resampling

Chapter Review

- Weighted least squares for unequal error variances
- Ridge regression for multicollinearity problem
- Robust regression for outliers / influential points
- Regression tree for nonparametric regression
- Evaluating precision in nonstandard situations using bootstrapping